

TAYLKOMB Rev B Engineering Audit and Auto-Routing Meta-Trigger Deep Research

[SALES13::APPLY + SALES13::DESIGN + PROMPTLIB::ROUTE + AUTO::DETECT+ROUTE active]

Scope and evidence base

This report synthesizes two tightly linked workstreams you provided: (a) the drawing-verified engineering audit for the six-part modular comb system (Main Comb, Wide Comb, Narrow Comb, Round Handle, Flat Handle, Double Handle) and (b) the “meta-trigger” layer intended to sit above your existing command map, SALES13 doctrine, and Prompt Library routing. [filecite\turn0file1\](#) [filecite\turn0file3\](#) [filecite\turn0file0\](#)

Because the original six drawing PDFs (e.g., [Main Comb.pdf](#), [Round Comb Handle.pdf](#)) are referenced but not directly available as standalone files in the tool layer here, the geometric facts in this report rely on your drawing-extraction tables and follow-on Rev B prep audit (treated as “source-of-truth” for dimensions), then cross-checked against market benchmarks, ergonomics guidance, and materials/DFM references from primary or first-party sources where available. [filecite\turn0file1\](#) [filecite\turn0file3\](#)

Drawing-verified geometry and system-level implications

Your drawing extraction shows unusually strong **cross-part standardization of the connector block**: repeated datums on all six parts include **61 mm**, **45.2 mm**, **5.08 mm**, and **~8.0–8.05 mm** features, with consistent **20.32 mm** depth appearing on the handle-side drawings (and closely matching ~20.35–20.79 mm values on comb-side drawings). This is a meaningful positive signal: a stable, shared interface datum stack is the single most important prerequisite for a durable modular tool family. [filecite\turn0file1\](#)

At the same time, the extracted overall lengths—**Main Comb ~228.81 mm**, comb attachments ~223 mm, and handles ~242–255 mm—produce an “assembled length inflation” problem when modules are combined. Your own audit computes assembled combinations in the **~405–424 mm** range (comb + handle – 61 mm overlap), placing the system at roughly **1.7–1.9×** the length of common single-piece pro combs. Whether that’s acceptable depends on the intended use case (e.g., long-hair sectioning vs. close-in cutting), but it is mechanically and ergonomically consequential because small increases in reach increase wrist moment and reduce precision in near-scalp maneuvers. [filecite\turn0file1\](#) [filecite\turn0file2\](#)

Three drawing-verified dimensions stand out as “high leverage” for Rev B because they couple directly to fatigue, breakage risk, and adoption:

- **Round handle tip diameter ~1.3 mm** (rat-tail) is a precision win but a durability risk in polymer if users apply lateral bending or leverage; several high-end tail combs solve the “thin + durable” problem by using a **metal tail** feature rather than pushing plastic tips extremely thin.

[filecite\turn0file1\](#) [1](#)

- **Double handle outer width ~27.2 mm** and prong cross-sections ~5.4–5.6 mm are likely to feel under-supported for sustained gripping, especially if the prongs are also very long (~192 mm).
□filecite□turn0file1□
- **Flat handle overall length ~255.6 mm** (with a long working body section) is outside the common “fast reposition” envelope for salon comb workflows, even before assembly length is considered.
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These are consistent with the Rev B prep audit’s headline concern that the system is “too long as a family,” with the most urgent redesign focus clustered around handle geometry, connector/seam zone robustness, and long-cantilever failure points. □filecite□turn0file3□

Market benchmarks and length-family calibration

The premium benchmark set you cited (and that can be verified directly from manufacturer pages) strongly supports your conclusion that **multiple distinct length families** exist in professional use—especially separating “close-in cutting control” from “long sectioning / detangling reach.”

On the long cutting / sectioning end, **YS Park** ² lists the 331 cutting comb at **9 inches / 229 mm**, explicitly pairing that length with fine cutting on longer hair and emphasizing gradually decreasing pitch (GDP) and heat resistance claims tied to ultem (PEI). ³ On the same brand’s tail-comb side, YS Park lists the 111 tail comb at **8.7 inches / 221 mm** and the 106 tail comb at **9.8 inches / 245 mm**, again reinforcing GDP and a “soap and water only” disinfection instruction. ⁴

For a second benchmark, **Sam Villa** ⁵’s Signature Series Long Cutting Comb lists **8.75 inches** overall (≈ 222 mm), and—critically for your seam strategy—claims “computerized molds eliminate all parting lines and guarantee perfect symmetry between each tooth,” framing snag-free glide as a premium expectation rather than a bonus feature. ⁶

For a third benchmark, **Hercules Sägemann** ⁷ emphasizes traditional manufacturing and finishing: ebonite (vulcanized natural hard rubber), “hand-sawn, handcut and hand-polished,” “perfectly rounded and burr-free teeth,” and explicit “chemical and heat-resistant” positioning. Regardless of material differences, the market signal is that finishing quality (no burrs, no sharp transitions, smooth tooth symmetry) is central to perceived value. ⁸

These benchmarks make your internal “two-family vs. single-family” decision point well-founded: forcing all modules into one long reach family simplifies visual cohesion, but it risks under-serving the “fast cutting comb” expectation. Your own version strategy memo explicitly frames this as a strategic fork (single long-length family vs. dual-family with a short precision set). □filecite□turn0file2□

Ergonomics implications that materially affect adoption

Two ergonomics anchor points are unusually applicable here because combs are, functionally, “precision hand tools” for much of their use:

- The **California Department of Industrial Relations** ⁹ ergonomics guidance for **precision tasks** calls for **single-handle tool diameters of 1/4 inch to 1/2 inch** (≈ 6.35 – 12.7 mm). ¹⁰

- **National Institute for Occupational Safety and Health** ¹¹ guidance repeats the same handle diameter range for precision tasks and distinguishes larger diameters for power tasks, reinforcing the principle that “effective diameter” should match grip mode. ¹²

This matters directly for your handle modules:

- The round handle’s “pencil-grip” region should be engineered so the **effective diameter where the fingers actually pinch** lands inside the precision band, even if upstream geometry is larger for stiffness and connector carry. If the taper passes through the 8–13 mm range at the index point, it will feel “tool-like” rather than “wand-like.” This is consistent with your CAD handoff guidance that precision grip effective diameter should sit in a narrow band (8–16 mm, nominal ~12 mm). ¹²
- The double handle’s very narrow outer width (~27.2 mm) is likely to land in a “no-man’s-land” between pinch control and palm support. If the intention is power leverage (twisting, tensioning), the effective grasp zone typically benefits from moving toward the larger “power-task” handle diameter range cited by NIOSH (1.25–2 inches = 31.75–50.8 mm), even if you implement that via a flattened or forked geometry rather than a round cylinder. ¹²

A second ergonomics implication is tooth length and overall height. Your drawing extraction flags a **~57.05 mm tooth length** and **~68.51 mm total height** for the Main Comb, which is atypically tall for a “cutting comb” positioning but plausible for detangling or textured-hair control. By contrast, the Sam Villa long cutting comb publishes wide/fine tooth lengths of roughly **4.5 inches / 4.25 inches** (≈ 114 mm / 108 mm) across the *toothed span*, but those are not “tooth depth” values; the key takeaway is that premium brands publish geometry in ways that relate to functional zones and glide, and that your “Main Comb” may need renaming or refocusing if tooth depth is intended for detangling rather than cutting. ⁶

Connector survivability, manufacturability, and materials under real salon chemistry

A modular interface for a styling tool faces three stresses simultaneously: (a) repeated mechanical cycling (swap-on/off), (b) hair product residue and debris ingress, and (c) chemical exposure either from professional disinfectants or from “soap and water” cleaning protocols.

Interface strategy for high-cycle, low-wobble performance

Your internal guidance consistently converges on an interface that is **anti-rotation + anti-wobble by geometry** and **retained by a low-strain detent**, rather than relying on a high-deflection snap arm as the primary load path. ¹³

This is strongly supported by classic plastics snap-fit guidance: **DuPont** ¹³’s engineering polymers design reference emphasizes uniform wall thickness and explicitly warns that allowable strain depends on whether parts are used once or frequently assembled/disassembled, providing suggested allowable strain values (including low allowable strain ranges for repeated-use snap features in common engineering polymers). ¹⁴

Manufacturability then dictates how tight you can realistically hold “no wobble” targets. Typical injection molding tolerance guidance notes that consumer-product tolerances often land around ± 0.1 mm, while tighter tolerances can be achieved at higher cost and with stricter process control. This supports your existing caution that “perfect on CAD” can still wobble after shrink variability and wear, especially if alignment surfaces are short or lightly constrained. ¹⁵

The resulting best-fit strategy for TAYLKOMB remains: long engagement geometry (your drawings already imply a generous 61 mm datum length), keyed anti-rotation surfaces, and a detent that is easy to replace or overbuild for cycle life—paired with a seam design that actively sheds hair. This also aligns with the market’s “no parting lines / no snagging” premium expectation stated by Sam Villa. ⁶

Disinfection stance and material selection

A critical strategic decision is whether you want to align with “soap and water only” guidance or typical salon immersion protocols. YS Park explicitly states across product pages that combs/brushes should be disinfected with **soap and water only**, which may reflect conservative material-care instructions rather than universal salon practice. ¹⁶

If you intend compatibility with common salon disinfectants, you must treat chemistry as a first-class requirement. The Barbicide label identifies quaternary ammonium chemistry (e.g., alkyl dimethyl benzyl ammonium chloride) as an active ingredient class, reinforcing that you are designing against surfactant + disinfectant exposure rather than “plain water.” ¹⁷

Material choice should therefore be framed as “mechanical + chemical + finishability” rather than resin marketing:

- POM (acetal) is frequently selected for low-friction moving contact and good dimensional stability; a technical plastics data reference summarizes POM as having excellent chemical resistance to many organics and resistance to several weak acids/alkalis, while stressing that specific chemical compatibility must be tested. ¹⁸
- For higher-end chemical/disinfectant regimes, specialty sulfone polymers (e.g., PPSU) are commonly benchmarked in medical-equipment housing contexts; Solvay’s disinfectant resistance study describes exposure methods including immersion and ESCR (referencing methods “similar to ASTM D543”) across multiple disinfectants, which is directly analogous to what you’d want for accelerated salon-chemistry validation. ¹⁹
- If PBT (e.g., Ultradur) is in consideration for dimensional stability and low water absorption, BASF positioning emphasizes chemical resistance and dimensional performance in medical component contexts, again suggesting the right evaluation lens is “exposure + stress + time,” not just datasheet properties. ²⁰

The practical implication for Rev B is that your “prototype-first” list should explicitly include: repeated connector cycling **after** disinfectant exposure, not just before. That is, you want to measure insertion/removal force and wobble growth across cycles in the presence of residue and after chemical soak—because environmental stress cracking and wear are often interaction effects (stress + chemical + time). ²¹

Finally, your rat-tail tip problem (1.3 mm) intersects directly with materials: if you want an ultra-thin precision tail **and** high durability, the market precedent is a plastic body paired with a **metal tail** element

(again: YS Park 132 explicitly references a “strong metal tail” for foil support). This supports your own option set of either thickening the plastic tip or engineering a metal insert as the precision/durability solution.

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AUTO::DETECT+ROUTE meta-trigger design and security research

Your missing layer—the “meta-trigger” that automatically applies SALES13 and routes into Prompt Library assets—maps cleanly onto a well-studied pattern in modern agent systems: a **router stage** that classifies intent, then retrieves the correct “expert” (prompt/workflow/agent) dynamically.

Alignment with your existing command architecture

Your unified command mapping explicitly defines top-level activation commands and doctrine layers (e.g., AGM, SALES13 commands, Prompt Library routing commands), establishing a structured command vocabulary suitable for machine-driven activation rather than manual recall. filecite turn0file0

Practically, AUTO::DETECT+ROUTE can be implemented as an orchestration wrapper that: (1) detects task type(s), (2) selects doctrine layer(s) like SALES13, (3) selects the Prompt Library target asset, (4) runs the execution prompt, and (5) returns an activation trace line. This is consistent with how “model routers” and “multi-expert routing” are commonly described in production LLM stacks. 22

Routing method that scales beyond keyword matching

Keyword/alias routing is a good baseline, but the research and practitioner guidance increasingly prefers **semantic search (embeddings)** as a robust alternative to pure classifier prompts or brittle keyword lists. An AWS guidance article on multi-LLM routing describes semantic search routing by embedding similarity between the incoming prompt and reference prompts representing task categories, which is directly analogous to routing into your parent titles / prompt registry. 23

A practical hybrid that fits your Prompt Library v2 constraints is:

- Use a lightweight signal extractor (keywords + a small intent classifier) for coarse category selection (e.g., “design critique,” “offer,” “pitch,” “EPUB,” “code review”).
- Within that category, use semantic similarity over your registry (titles, aliases, component types) to pick the exact parent title and asset.

This reduces false positives (keyword collisions) and allows “natural language” matching without requiring the user to remember command syntax—matching the explicit goals of ZERO-COMMAND::ON. 23

Security and prompt-injection implications of auto-detect layers

A meta-trigger that auto-activates doctrine and routes across assets is also a **larger attack surface** if it processes untrusted external text (e.g., pasted emails, web content, supplier messages). Prompt injection is now widely cataloged as a core LLM application risk by OWASP 24’s LLM Top 10 (LLM01), and NIST 25 explicitly calls out prompt injection as a risk case in generative AI contexts within its AI RMF documentation.

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In practice, this means AUTO::DETECT+ROUTE should adopt “defense-in-depth” patterns if it will ever ingest third-party content. **Anthropic** ²⁷’s public work on prompt injection defenses emphasizes that no agent is immune and that even low attack success rates can remain meaningful at scale—supporting the need for strict separation between “data” (what the user pasted) and “instructions” (what the router obeys). ²⁸

A concrete, research-backed security posture for your router layer looks like this:

- Treat all external pasted text as untrusted “data,” never as instructions; only the command map + routing guide should define execution rules. This aligns with OWASP’s framing of prompt injection as unintended behavior change via crafted inputs. ²⁹
- Keep AUTO::DETECT+ROUTE’s output machine-readable internally (classification + selected asset + confidence), and only then render the user-facing response. This mirrors common router implementation patterns that parse structured outputs. ³⁰
- Add an explicit “high-risk” gate for actions that could cause harm (e.g., instructions that look like malware or credential theft) so the router cannot be tricked into activating “hostile code review” or sensitive workflows merely by keyword. This follows the general secure design posture recommended in modern AI security frameworks and threat discussions. ³¹

Prior art note for “modular comb” novelty claims

Because your engineering audit references “patent-pending,” it is worth noting (purely from a research perspective, not legal advice) that **modular/multifunction comb concepts exist in published patent literature**, including designs describing selectable or attachable tooth bodies. This does not invalidate your work, but it does raise the bar on claim novelty and emphasizes the need for a patent professional to position the differentiators (e.g., connector geometry, salon workflow rationale, durability targets, seam anti-snag design). ³²

¹ ¹² YS Park 132 Tail Comb (Extra Long) - 9.4"

<https://ysparkusa.com/products/ys-park-tail-comb-132?srsId=AfmBOorXhZq8mgagt3FfZr3sIyBuVo-5iwdsIT1ITlxxTSFnTGGSGBy5>

² ¹² A Guide to Selecting Non-Powered Hand Tools

https://www.cdc.gov/niosh/docs/2004-164/pdfs/2004-164.pdf?utm_source=chatgpt.com

³ ¹⁶ YS Park 331 Fine Cutting Comb (Extra Super Long)

<https://ysparkusa.com/products/ys-park-cutting-comb-331?srsId=AfmBOordPB6sbhUv6MJH4PiK7E8DRuTwSCv2ctyys7072tgglQCQyTd>

⁴ ¹³ YS Park 111 Tail Comb - 8.7"

https://ysparkusa.com/products/ys-park-tail-comb-111?srsId=AfmBOoprgejq-wysn6_dHq2aIpS9pawYRn69lUE8G2KHnwgHLOVXxL7D&utm_source=chatgpt.com

⁵ ¹⁴ General Design Principles for DuPont Engineering Polymers

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⁶ ²⁴ Signature Series Long Cutting Comb | Sam Villa Combs

https://www.samvilla.com/products/long-cutting-comb?srsId=AfmBOor6_A6PT9hn5umZZpjZLVYZ8l1K1iocV3FVSglA5-Wxx4d1v9Fo

⁷ ²⁹ LLM01:2025 Prompt Injection - OWASP Gen AI Security Project

https://genai.owasp.org/llmrisk/llm01-prompt-injection/?utm_source=chatgpt.com

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